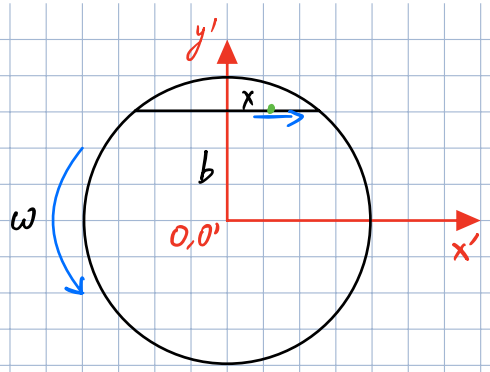
3.3



直角坐标系 $O'x'y'$ 随圆盘转动.

$$\vec{r} = \vec{r}_{O'} + \vec{r}' = x\vec{i}' + b\vec{j}'$$

$$\vec{v} = \dot{\vec{r}} = \dot{x}\vec{i}' + x\dot{\vec{i}}' + b\dot{\vec{j}}'$$

$$= u\vec{i}' + x\omega\vec{j}' - b\omega\vec{i}'$$

$$= (u - b\omega)\vec{i}' + x\omega\vec{j}'$$

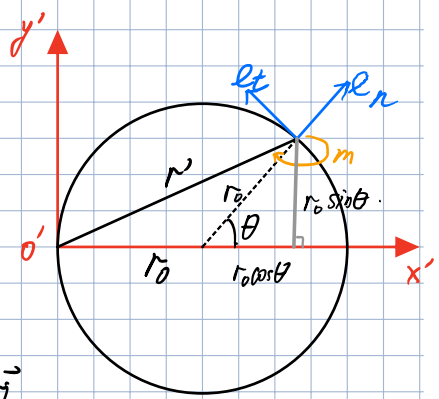
$$\frac{d\vec{i}'}{dt} = \vec{\omega} \times \vec{i}' = \omega\vec{j}'$$

$$\frac{d\vec{j}'}{dt} = \vec{\omega} \times \vec{j}' = -\omega\vec{i}'$$

$$\vec{a} = (u - b\omega)\omega\vec{j}' + u\omega\vec{j}' - x\omega^2\vec{i}'$$

$$= (2u - b\omega)\omega\vec{j}' - x\omega^2\vec{i}'$$

3.12



解: $\vec{r}' = r_0(1+\cos\theta)\vec{i} + r_0\sin\theta\vec{j}$

$\vec{v} = -r_0\dot{\theta}\sin\theta\,\vec{i} + r_0(1+\cos\theta)\omega\vec{j}$

$+ r_0\dot{\theta}\cos\theta\,\vec{j} - r_0\omega\sin\theta\,\vec{i}$

$= -r_0\sin\theta(\dot{\theta}-\omega)\vec{i} + r_0(1+\cos\theta+\dot{\theta}\cos\theta)\vec{j}$

$\vec{a} = -r_0\dot{\theta}\cos\theta(\dot{\theta}-\omega)\vec{i} - r_0\sin\theta\cdot\ddot{\theta}\vec{i} - r_0\sin\theta(\dot{\theta}-\omega)\omega\vec{j}$

$+ r_0(-\sin\theta\cdot\dot{\theta} + \ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta)\vec{j} - r_0\omega(1+\cos\theta+\dot{\theta}\cos\theta)\vec{i}$

$= r_0\vec{i}(-\dot{\theta}^2\cos\theta + \dot{\theta}\omega\cos\theta - r_0\sin\theta\cdot\ddot{\theta} - \omega - \omega\cos\theta - \omega\dot{\theta}\cos\theta)$

$+ r_0\vec{j}(-\dot{\theta}\omega\sin\theta + \omega^2\sin\theta - \dot{\theta}\sin\theta + \ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta)$